

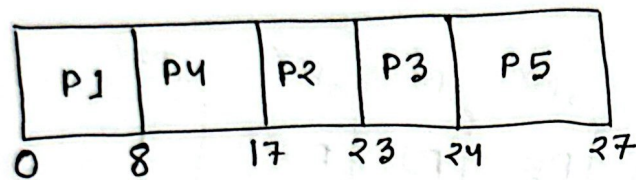
1. Given the arrival times, burst times and priority values for the processes below. Draw gantt charts and calculate the average waiting time and turnaround time using the scheduling algorithms given below:
- I. First Come First Serve
 - II. Shortest Remaining Job First
 - III. Preemptive Priority
 - IV. Round Robin (time quantum: 5)

Process	Arrival Time	Burst Time	Priority
P1	0	8	4
P2	2	6	1
P3	2	1	2
P4	1	9	2
P5	3	3	3

CPU scheduling

① i) First come First serve (FCFS)

Process	Arrival time	Burst time	Priority	Waiting time	Turnaround time
P1	0	8	1	0	$8 - 0 = 8$
P2	2	6	1	15	$23 - 2 = 21$
P3	2	1	2	21	$24 - 2 = 22$
P4	1	9	2	7	$17 - 1 = 16$
P5	3	3	3	21	$27 - 3 = 24$

Waiting time

P1: $0 - 0 = 0$

P4: $8 - 1 = 7$

P2: $17 - 2 = 15$

P3: $23 - 2 = 21$

P5: $24 - 3 = 21$

$$\text{average waiting time} = \frac{0 + 7 + 15 + 21 + 21}{5} = 12.8$$

$$\text{Turnaround time} = \text{termination} - \text{arrival time}$$

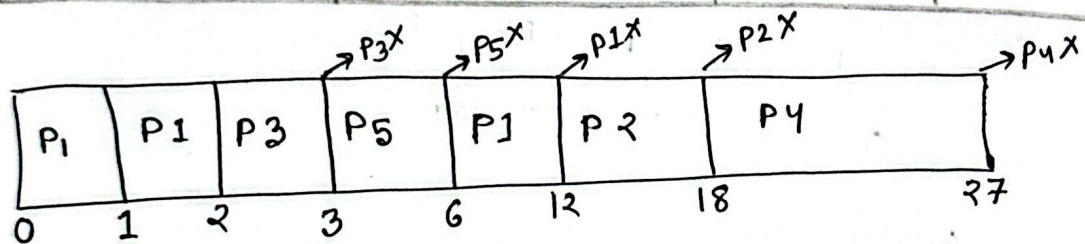
$$= (8 - 0) + (17 - 1) + (23 - 2) + (24 - 2) + (27 - 3)$$

$$= 91$$

$$\text{average turnaround time} = \frac{91}{5} = 18.2$$

(ii) Shortest Remaining Job First (SRJF)

Process	Arrival time	Burst time	Waiting time	Turnaround time	Response time
P1	0	8 8	$12 - 8 = 4$	$12 - 0 = 12$	0
P2	2	6	$16 - 6 = 10$	$18 - 2 = 16$	$12 - 2 = 10$
P3	2	1	$1 - 1 = 0$	$3 - 2 = 1$	$2 - 2 = 0$
P4	1	9	$26 - 9 = 17$	$27 - 1 = 26$	$18 - 1 = 17$
P5	3	3	$3 - 3 = 0$	$6 - 3 = 3$	$3 - 3 = 0$



Queue:

0 $\rightarrow$ (P1, 8)

1 $\rightarrow$ (P1, 7), (P4, 9)

2 $\rightarrow$ (P1, 6), (P4, 9), (P2, 6), (P3, 1)

3 $\rightarrow$ (P1, 6), (P4, 9), (P2, 6), (P5, 3)

6 $\rightarrow$ (P1, 6), (P2, 6), (P4, 9)

12 $\rightarrow$ (P2, 6), (P4, 9)

18 $\rightarrow$ (P4, 9)

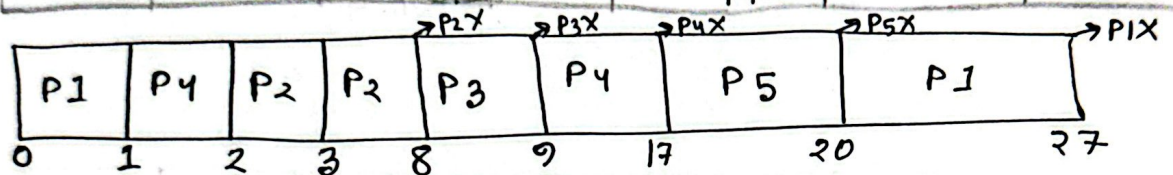
Waiting time = turnaround time - Burst time

average waiting time = $(4 + 10 + 0 + 17 + 0) / 5 = 6.2$

average turnaround time = $(12 + 16 + 1 + 26 + 3) / 5 = 11.6$

(iii) Preemptive Priority

Process	Arrival time	Burst time	Priority	Turnaround time	Waiting time	Response time
P1	0	8 0	4	$27 - 0 = 27$	$27 - 8 = 19$	0
P2	2	6 0	1	$8 - 2 = 6$	$6 - 6 = 0$	$2 - 2 = 0$
P3	2	20	2	$9 - 2 = 7$	$7 - 1 = 6$	$8 - 2 = 6$
P4	1	9 80	2	$17 - 1 = 16$	$16 - 9 = 7$	$1 - 1 = 0$
P5	3	30	3	$20 - 3 = 17$	$17 - 3 = 14$	$17 - 3 = 14$
				average = 14.6	average = 9.2	



Queue:

0: (P1, 8)

1: (P1, 7), (P4, 9)

2: (P1, 7), (P4, 8), (P2, 6), (P3, 1)

3: (P1, 7), (P4, 8), (P2, 5), (P3, 1), (P5, 3)

8: (P1, 7), (P3, 1), (P4, 8), (P5, 3)

9: (P1, 7), (P4, 8), (P5, 3)

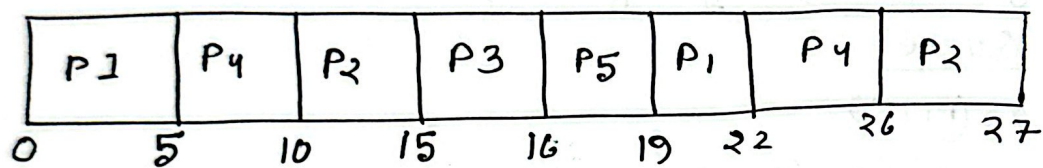
17: (P1, 7), (P5, 3)

20: (P1, 7)

27: {}

(iv) Round Robin (time quantum : 5)

Process	Arrival time	Burst time	Turnaround time	Waiting time	Response time
P1	0	8	$22 - 0 = 22$	$22 - 8 = 14$	0
P2	2	6	$27 - 2 = 25$	$25 - 6 = 19$	$10 - 2 = 8$
P3	2	2	$16 - 2 = 14$	$14 - 1 = 13$	$15 - 2 = 13$
P4	1	5	$26 - 1 = 25$	$25 - 9 = 16$	$5 - 1 = 4$
P5	3	3	$19 - 3 = 16$	$16 - 3 = 13$	$16 - 3 = 13$
			average = 20.4	average = 15	

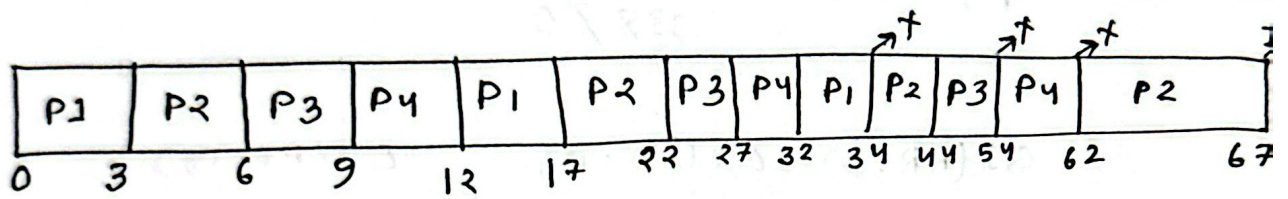


2. Given the arrival times and burst times for the processes below, draw a gantt chart using multilevel feedback queue scheduling algorithm showing the states of the ready queues of different levels. Then calculate the average waiting time, response time and turnaround time. There are 3 queues with decreasing priority all using round robin scheduling algorithm internally. **Q1 (quantum: 3, priority:0), Q2 (quantum:5, priority:1), Q3 (quantum:10, priority:2)**

Process	Arrival Time	Burst Time
P1	0	10
P2	2	23
P3	5	18
P4	7	16

② MLFQ

Process	Arrival time	Burst time	Turnaround time	Waiting time	Response time
P1	0	10	34 - 0 = 34	34 - 10 = 24	0
P2	2	3	65 - 2 = 63	63 - 3 = 60	3 - 2 = 1
P3	5	18	54 - 5 = 49	49 - 18 = 31	6 - 5 = 1
P4	7	16	62 - 7 = 55	55 - 16 = 39	9 - 7 = 2
			average = 50.75	average = 34	average = 1



Q1 (RR: 3)	Q2 (RR: 5)	Q3 (RR: 10)
0: P1	3: P1	17: P1
2: P2	6: P1, P2	22: P1, P2
5: P3	9: P1, P2, P3	27: P1, P2, P3
7: P4	12: P1, P2, P3, P4	32: P1, P2, P3, P4
12: { }	17: P2, P3, P4	34: P2, P3, P4
	22: P3, P4	44: P3, P4, P2
	27: P4	54: P4, P2
	32: { }	62: P2
		67: { }

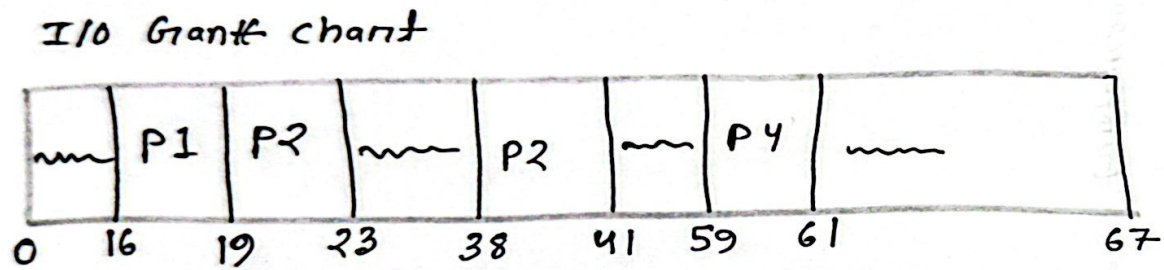
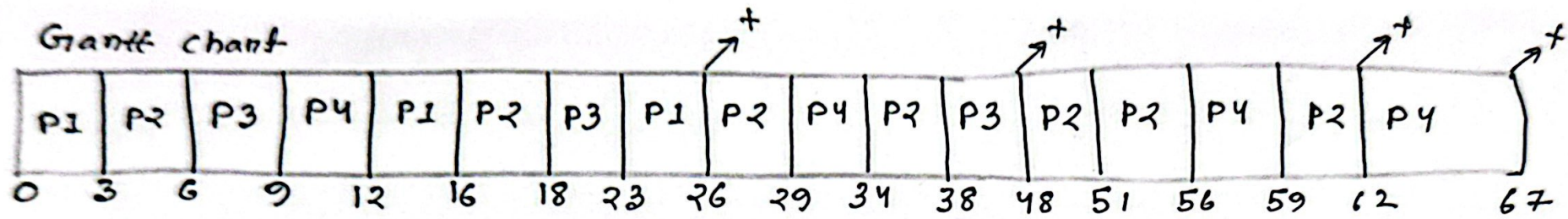
3. Given the arrival time and burst time for the processes below, draw a gantt chart using multilevel feedback queue scheduling algorithm showing the states of the ready queues of different levels. Then calculate the average waiting time, response time and turnaround time. There are 3 queues with decreasing priority all using round robin scheduling algorithm internally. **Q1 (quantum: 3, priority:0), Q2 (quantum:5, priority: 1), Q3 (quantum:10, priority:2)**. In case of I/O request, a process will be promoted one

Process	Arrival Time	Burst Time	I/O Request
P1	0	10	3s IO after 7s CPU
P2	2	23	4s I/O after 5s CPU 3s I/O after 12s CPU
P3	5	18	N/A
P4	7	16	2s I/O after 11s CPU

level upon I/O completion.

Process	Arrival time	Burst time	I/O Request
P1	0	10 30 7 40	3s I/O after 7s CPU
P2	2	23 20 18 15 14 8 30 5 20 12 8 7 40	4s I/O after 5s CPU 3s I/O after 12s CPU
P3	5	18 18 100	N/A
P4	7	16 13 8 5 0 11 8 30	2s I/O after 11s CPU

Q1 (RR:3)	Q2 (RR:5)	Q3 (RR:10)	I/O
0: P1	3: P1	34: P4	16: P1
2: P2	6: P1, P2	38: P4	19: { }
5: P3	9: P1, P2, P3	23: P3	18: P2
7: P4	12: P1, P2, P3, P4	34: P3, P4	23: { }
12: { }	16: P2, P3, P4	38: P3, P4	38: P2
19: P1	18: P3, P4	48: P4	41: { }
23: P1, P2	23: P4	51: P4	59: P4
26: P2	26: P4, P1	56: P4, P2	61: { }
29: { }	29: P4, P1, P2	59: P2	
41: P2	34: P1, P2	62: P2 { }	
51: { }	38: { }		
61: P4	51: P2		
	56: { }		
	61: P4		
	67: { }		



Process	Turnaround time	Waiting time	Response time
P1	$26 - 0 = 26$	$26 - 10 = 16 - 3 = 13$	0
P2	$62 - 2 = 60$	$60 - 23 = 37 - 7 = 30$	$3 - 2 = 1$
P3	$48 - 5 = 43$	$43 - 18 = 25 - 4 = 21$	$6 - 5 = 1$
P4	$67 - 7 = 60$	$60 - 16 = 44 - 2 = 42$	$9 - 7 = 2$
	average = 47.25	average = 27.5	average = 1

waiting time = turnaround time - Burst time - I/O